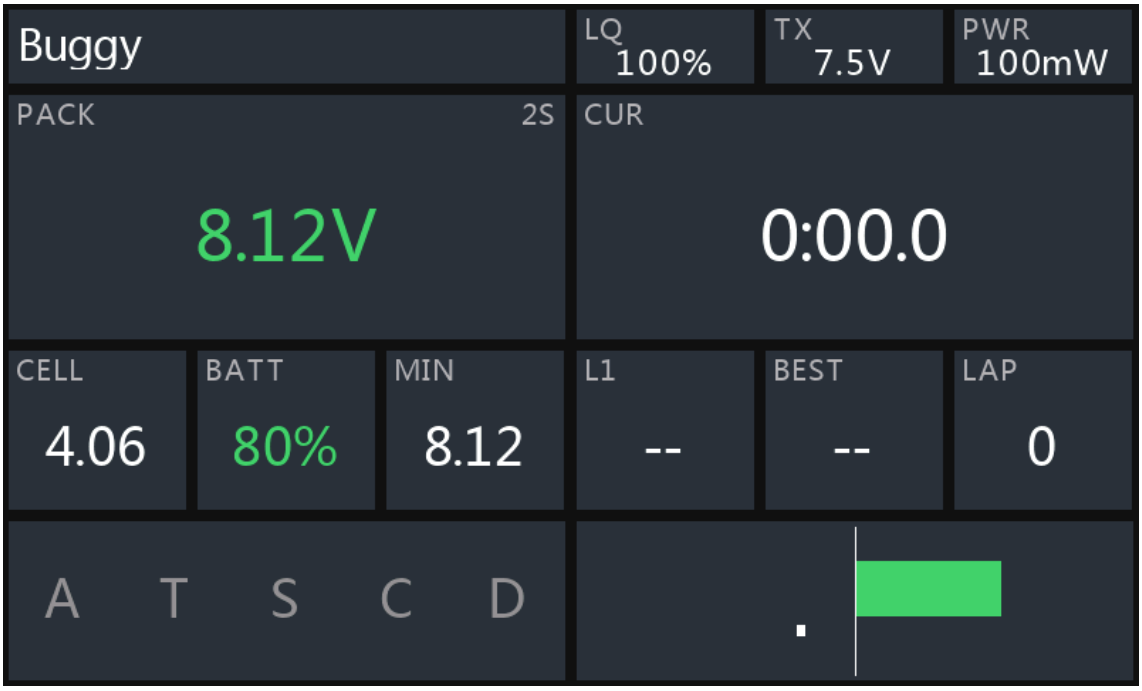
# Trackside

Race dashboard and drive aids
for the FrSky Velox VX10




*The Trackside dashboard on a VX10*

**Setup guide** | Version 0.1.0 | 26 October 2026

# Before you start

This guide sets up Trackside on a car model, from a fresh model through to the dashboard. It takes about ten minutes. Every screenshot is from the VX10.

| You need | |
|---|---|
| Radio | FrSky Velox VX10 with Ethos 26.2 or later |
| Software | Trackside 0.1.0 (the **trackside** folder) |
| Model | A car model. Section 2 creates one with the Ethos wizard. |

## How Trackside connects to your car

Trackside adds these items to your radio:

| Part | What it is | What it does |
|---|---|---|
| **Throttle Aid** and **Steer Aid** | Lua sources | Each outputs a multiplier for one channel. Your mixes keep the normal stick line and add one **Multiply** line with the aid after it. |
| **Trackside** | Telemetry sensor | Created automatically. Reads 1 while Trackside runs. A logic switch built on it turns the aid lines off if Trackside stops or is missing. |
| **Trackside** | System tool | In the System menu. Holds all Trackside settings and **Check setup**, which proves the aid mixes work. |
| **Trackside** | Widget | The race dashboard. |

Because the stick line stays in place, the car is always driven by your trigger and wheel. The aids can only reduce them. If the Trackside scripts are missing, the safety switch from section 4c turns the aid lines off and the car drives on the plain stick. If Trackside hits an error, the aids let the stick straight through.

**About this version.** The aids are set up and connected, but in version 0.1.0 they pass the sticks through unchanged. Setting them up now means nothing else needs to change when the aids arrive in an update.

**Safety.** Make mixer changes on the bench with the car's motor disconnected or the radio's RF output off. Check failsafe still works afterwards.

# 1. Install Trackside

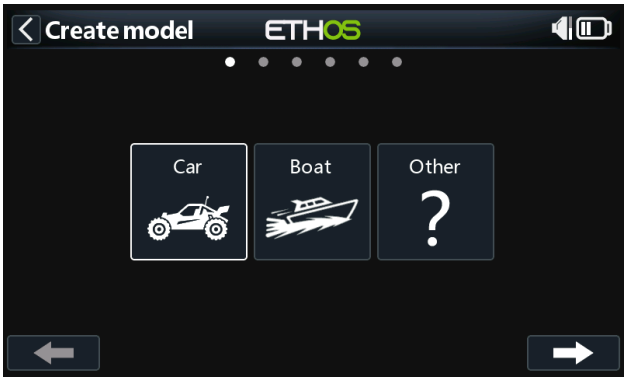
1. Connect the VX10 to your computer by USB and choose **USB storage** on the radio.
2. Copy the **trackside** folder into the radio's **SCRIPTS** folder, so you have **SCRIPTS/trackside/main.lua**.
3. Eject the radio drive safely and unplug it. The radio loads Trackside on its next start.

> Your settings are saved in **SCRIPTS/trackside.user**, a separate folder. Updating Trackside replaces only **SCRIPTS/trackside**, so your settings are kept.

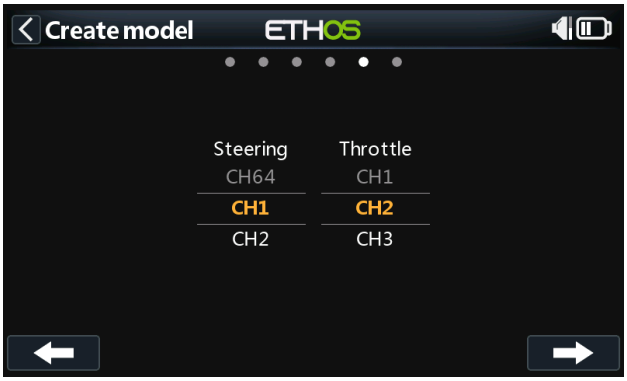
# 2. Create a car model

Skip this if you already have a car model with Steering on CH1 and Throttle on CH2. Otherwise create one with the Ethos wizard: **Model > Model select > +**.

1. Choose **Car**.
2. Step through the engine, steering and brake pages. The defaults are fine for most cars.
3. On the channel page, keep **Steering = CH1** and **Throttle = CH2**. The rest of this guide uses these.
4. Give the model a name and finish the wizard.




*Choose Car*




*Steering CH1, Throttle CH2*

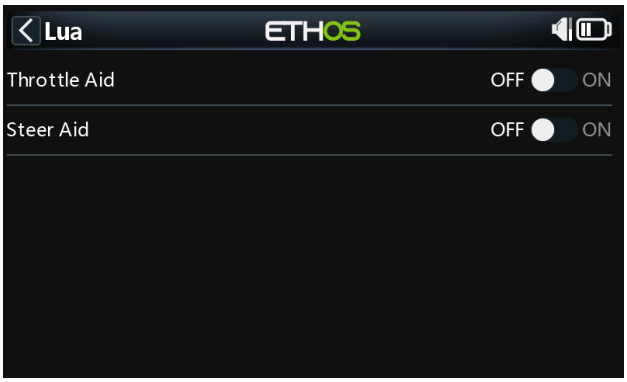
# 3. Turn on Throttle Aid and Steer Aid

Ethos only runs a Lua source, and only lists it when you pick a mix source, after you turn it on for the model. Do this once per model.

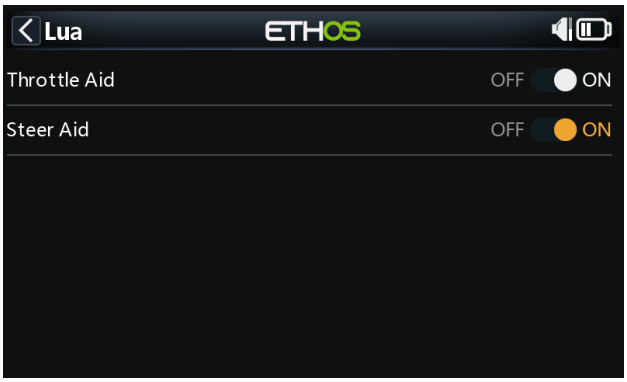
1. Open **Model** and swipe to the second page.
2. Tap **Lua**.
3. Turn **Throttle Aid** and **Steer Aid** ON. Trackside then also creates its **Trackside** telemetry sensor, used in section 4c.




*Model menu, page 2: the Lua page*




*Both start OFF*

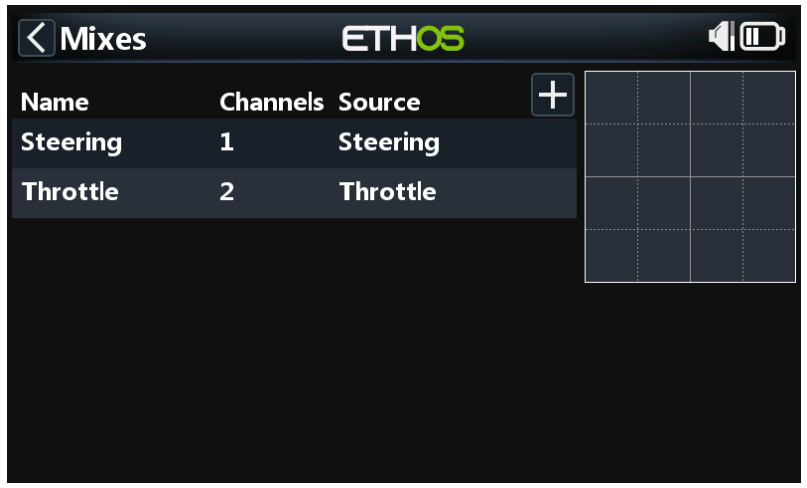



*Turn both ON*

# 4. Add the aid mixes

The car wizard made two mixes: **Steering** (CH1) and **Throttle** (CH2). **Leave them exactly as they are.** You add one more mix after each.

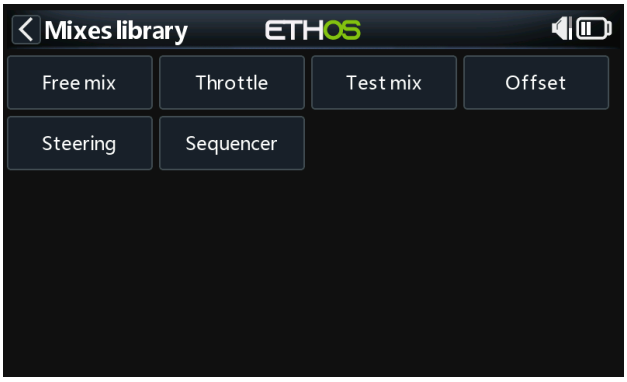
> **Don't** change the source of the existing Steering or Throttle mix to an aid. The channel would then follow the aid's fixed value instead of your stick, and nothing would move.



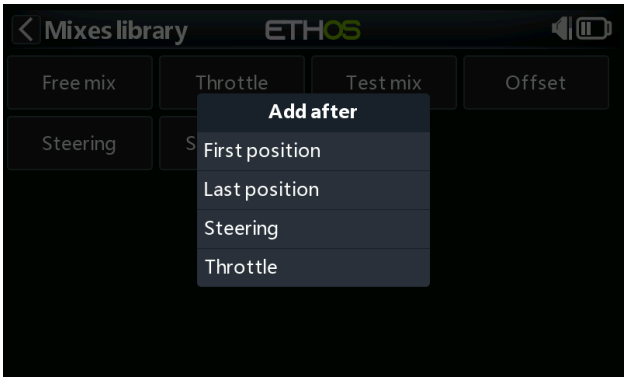

*Model > Mixes: the two mixes from the wizard*

## 4a. Throttle

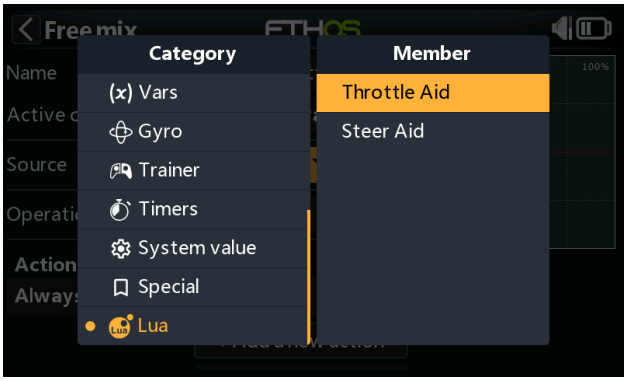
1. In **Mixes**, tap **+**, choose **Free mix**, then **Add after: Throttle**.
2. Optional: set **Name** to *Throttle Aid* so it's easy to spot in the list.
3. Tap **Source**, scroll to the **Lua** category and pick **Throttle Aid**.
4. Set **Operation** to **Multiply**.
5. Scroll down and set **Output** to **CH2 (Throttle)**. A new Free mix starts with Output = **---** and does nothing until you set it.



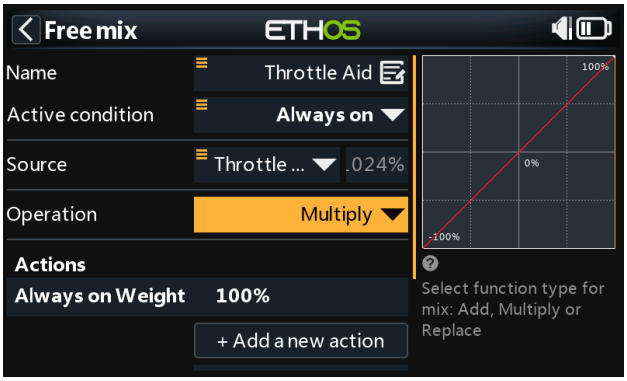

*+ opens the mix library: choose Free mix*




*Add after: Throttle*




Source: Lua > Throttle Aid




Operation: Multiply

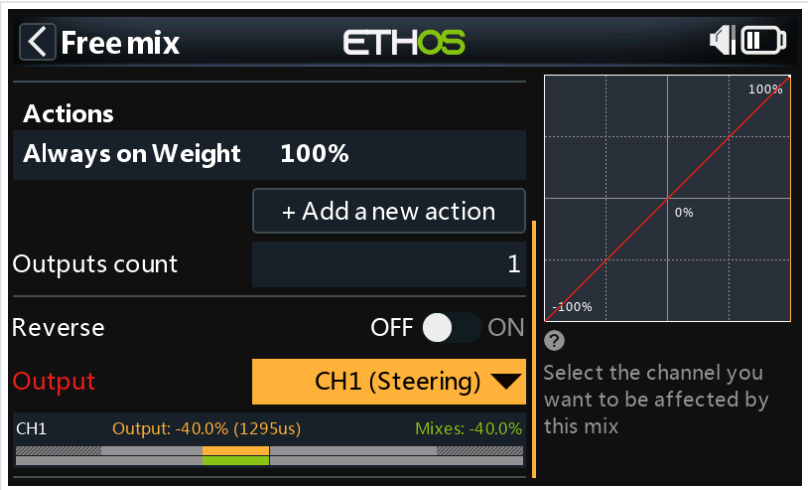



*Output: CH2. The bar shows the result: throttle 60% gives CH2 60%.*

> The grey box next to Source shows about **1024%**. That's Ethos showing the aid on its internal scale, and it's expected. What counts is the channel output.

## 4b. Steering

Repeat the same steps with these changes: **Add after: Steering**, source **Lua > Steer Aid**, Operation **Multiply**, Output **CH1 (Steering)**.



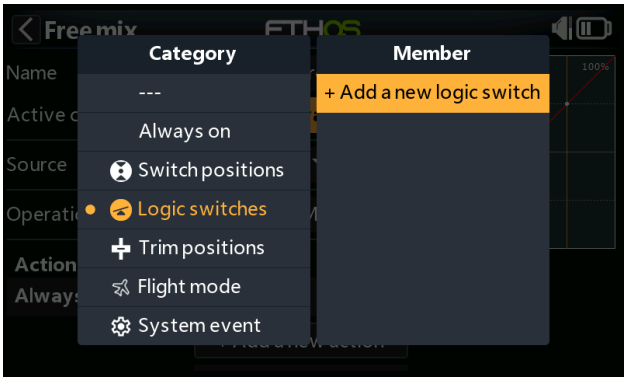

*Steer Aid on CH1: wheel at -40% gives CH1 -40%*

## 4c. Add the safety switch

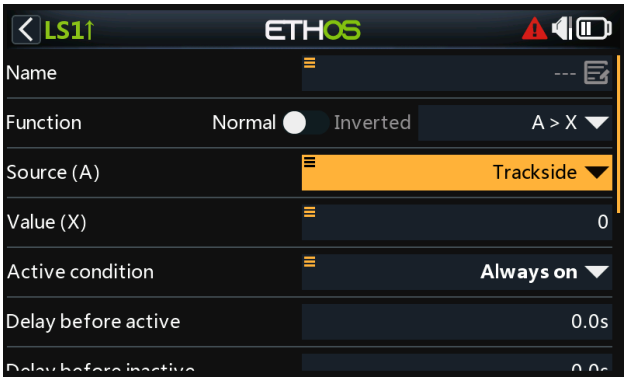
This makes the two aid lines run only while Trackside is loaded. You create one logic switch and use it as the **Active condition** of both aid lines.

> **Don't skip this.** If the Trackside scripts are missing (not copied, a new SD card, a failed update) or stops, Ethos treats each aid as zero or holds its last value. An aid line without the safety switch then multiplies its channel by zero: **no steering and no throttle**. With the switch, the aid lines turn off and the car drives on the plain stick.

1. Open the **Throttle Aid** mix: tap it, then **Edit**.
2. Tap **Active condition**, choose **Logic switches**, then **+ Add a new logic switch**.
3. Set **Function** to **A > X** and **Source (A)** to **Telemetry > Trackside**. Leave **Value (X)** at 0. The title **LS1** turns green while Trackside is running.
4. Tap **<** to go back. The mix's Active condition now reads **LS1**.
5. Open the **Steer Aid** mix and set its **Active condition** to **Logic switches > LS1**.




*Active condition: Logic switches > + Add a new logic switch*



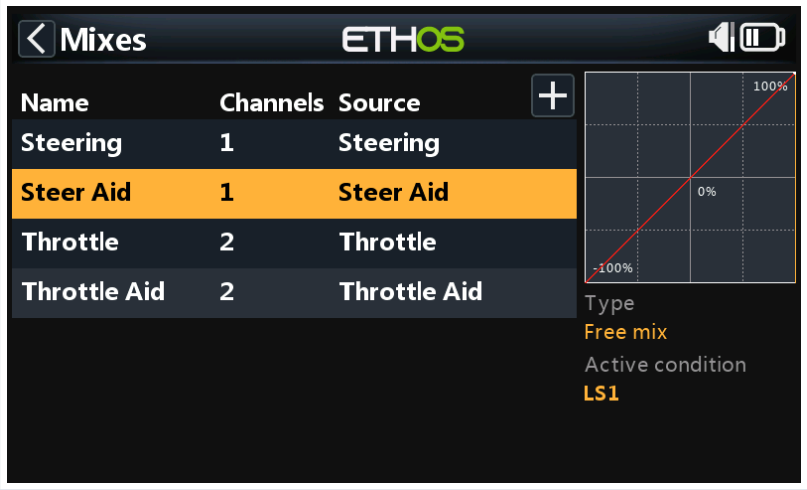

*LS1: A > X, Source (A) Trackside sensor, Value (X) 0*




*The aid line with Active condition LS1*

### 4d. Check the order

Each stick mix must be followed by its aid. An aid line placed **above** its stick mix does nothing. Use exactly one aid line per channel: a second Multiply line with the same aid applies it twice.




*Finished: Steering then Steer Aid (CH1), Throttle then Throttle Aid (CH2), both aids on LS1*

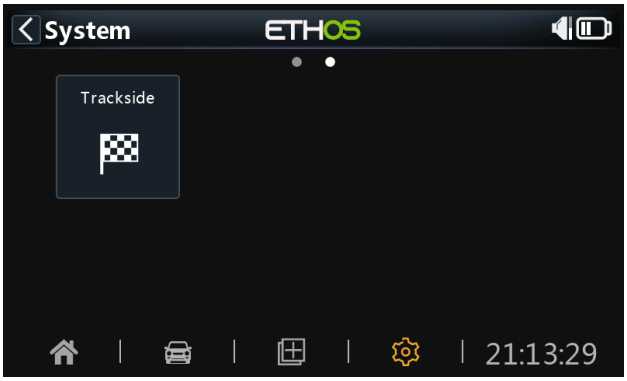
| Channel | Mix | Source | Operation | Active condition |
|---|---|---|---|---|
| CH1 | Steering | Analogs > Steering (the wheel) | Add | Always on |
| CH1 | Steer Aid | Lua > Steer Aid | Multiply | LS1 |
| CH2 | Throttle | Analogs > Throttle (the trigger) | Add | Always on |
| CH2 | Throttle Aid | Lua > Throttle Aid | Multiply | LS1 |

To reverse a channel, use **Model > Channels**. The wizard's stick mixes have no reverse option, and a reverse on the aid line flips the whole channel.

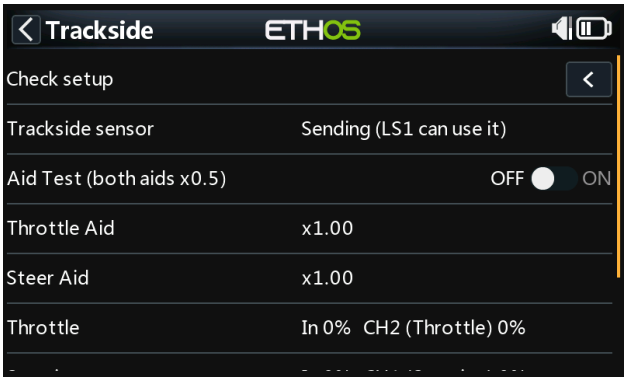
# 5. Check it works

In this version the aids pass the sticks through at x1.0, so a working setup looks the same as no setup. The **Aid Test** in the Trackside tool proves the mixes work: it sets both aids to x0.5, so CH1 and CH2 should move only half as far.

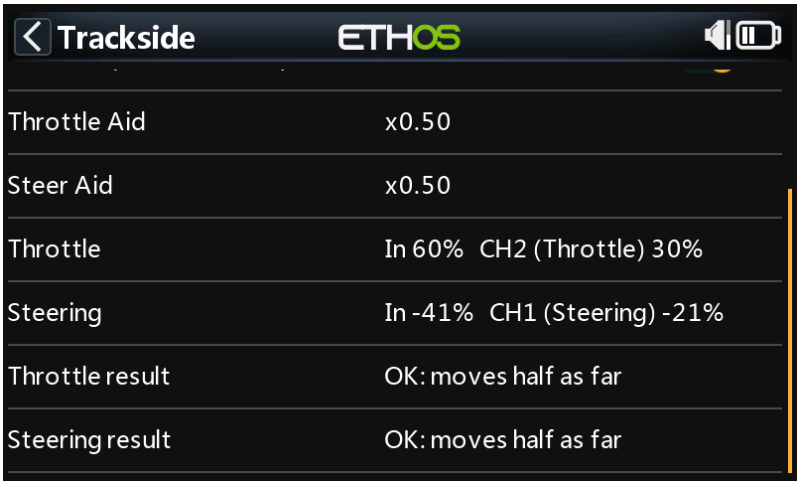
1. Open **System** (the gear in the bottom bar), swipe to the second page and tap **Trackside**.
2. Tap **Check setup**. **Trackside sensor** should read **Sending**, and both aids **x1.00**.
3. Turn **Aid Test** ON, then pull the trigger and turn the wheel past 20%.
4. Scroll down. Both results should read **OK: moves half as far**.
5. Leave the page with the back button. The Aid Test switches off by itself (also after 60 seconds).




*System, page 2: the Trackside tool*




*Check setup, Aid Test OFF*



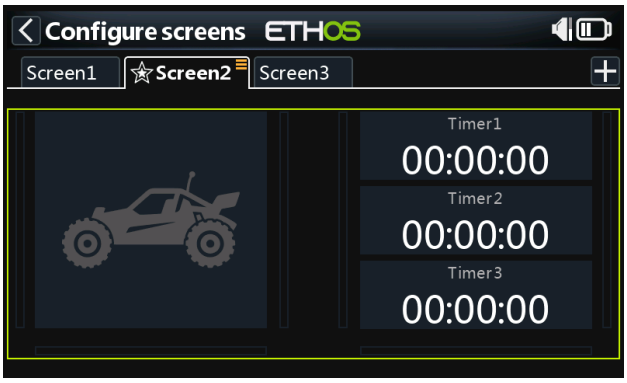

*Aid Test ON: trigger 60% gives CH2 30%, both results OK*

| Result | Meaning and fix |
|---|---|
| OK: moves half as far | The aid line works. |
| Not working: aid line missing or off | The channel ignored the aid. Check the aid line exists, its Output is the right channel, it sits below the stick mix, and LS1 is green (section 4c). |
| Wrong: set Operation Multiply | The aid line is there but changes the channel the wrong way, usually Operation **Add**. Set it to **Multiply**. |
| Throttle Aid: Not running | Turn the aid on in **Model > Lua** (section 3). |

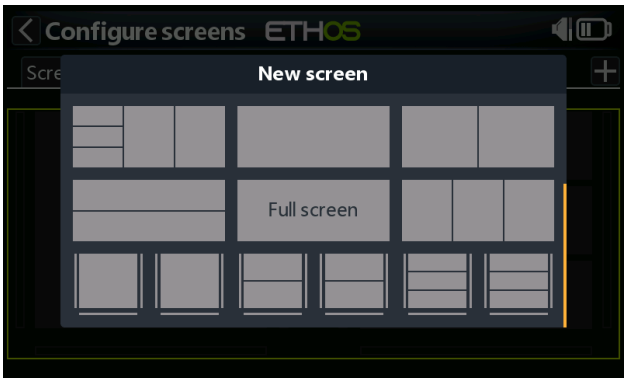
The check compares the channel to the input. If your aids drive other channels than CH1 and CH2, set **Throttle Channel** and **Steering Channel** on the tool's Inputs page.

# 6. Add the dashboard

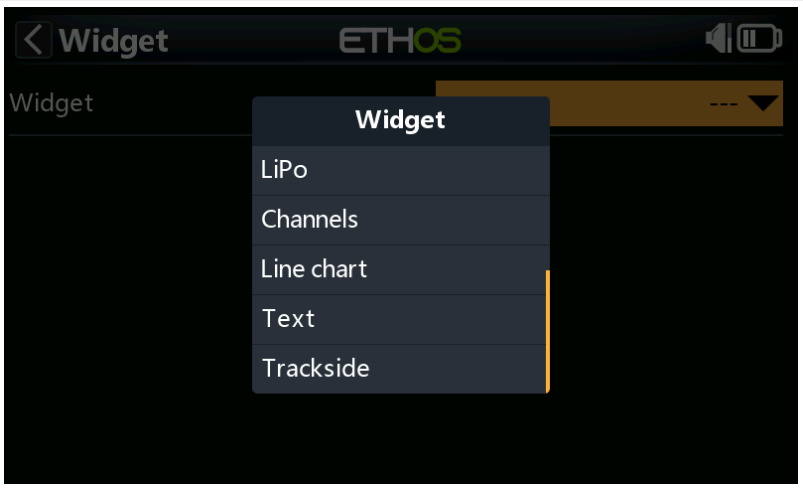
1. Tap the **Screens** icon in the bottom bar (the square with a +).
2. Tap **+** at the top right to add a screen.
3. Choose **Full screen**. Trackside also works in other layouts; in full screen it adds its own header with model name, link quality, radio battery and transmit power.
4. Tap the empty widget area, then pick **Trackside** from the widget list.
5. Go back to the home screen and swipe to the new screen.




*Screens: tap + to add a screen*



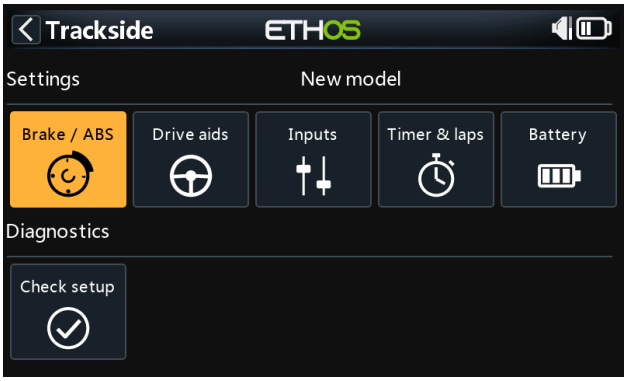

*Choose Full screen*



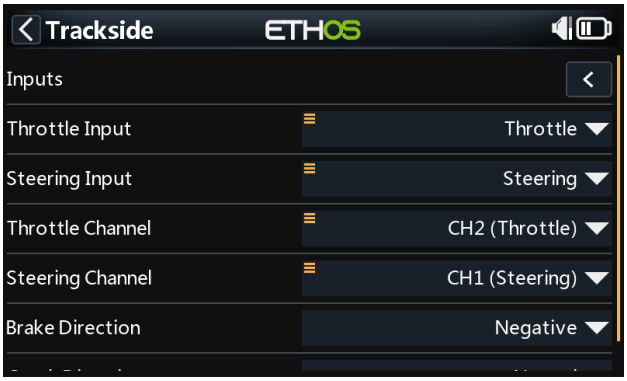

*Pick Trackside from the widget list*

# 7. Trackside settings

All settings are in the Trackside tool: **System**, second page, **Trackside**. Tap a tile to open its page; the back button returns to the menu. Settings are saved for the current model (its name is shown at the top) when you leave a page.




*The Trackside tool: settings and diagnostics*

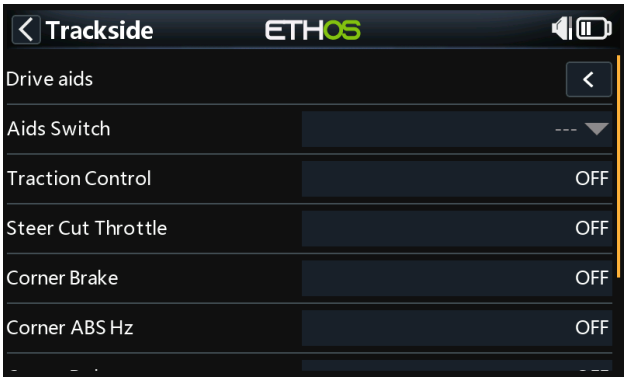



*Inputs*

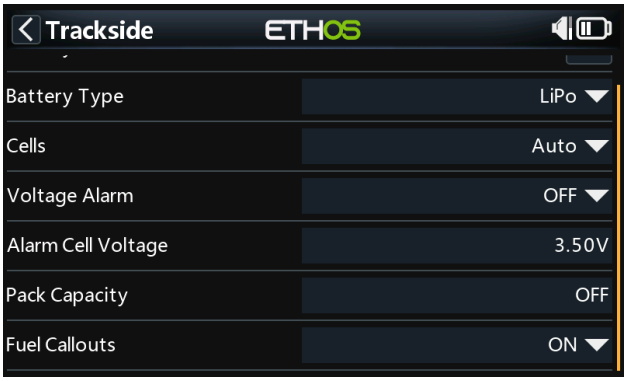
## Check these first

| Setting | Page | What to set |
|---|---|---|
| Throttle Input | Inputs | The control the aids read for throttle. Default: the VX10 trigger (Throttle). |
| Steering Input | Inputs | The control the aids read for steering. Default: the VX10 wheel (Steering). |
| Brake Direction | Inputs | Which way the trigger brakes. Pull the brake and watch the dashboard bar: it should turn red. |
| Battery Type, Cells | Battery | Match your packs (LiPo or LiHV). Cells = Auto detects the cell count. |
| Pack Capacity | Battery | Your pack size in mAh. Only needed if the car sends mAh used: fuel is then counted by mAh instead of estimated from voltage. |
| Fuel Callouts | Battery | Says "Fuel 50 percent" and so on at 50, 30, 20 and 10%. ON by default. |
| Voltage Alarm, Alarm Cell Voltage | Battery | Turn the low-voltage alarm on and choose the per-cell warning level. |

**Aids Switch** (top of the Drive aids page): pick a switch position, for example **SA↑**. While it's in that position the aids work; in any other position every aid lets the sticks straight through. Leave it at **---** to have the aids always on. The LS1 safety switch from section 4c is still needed: it covers the case where Trackside isn't running at all.



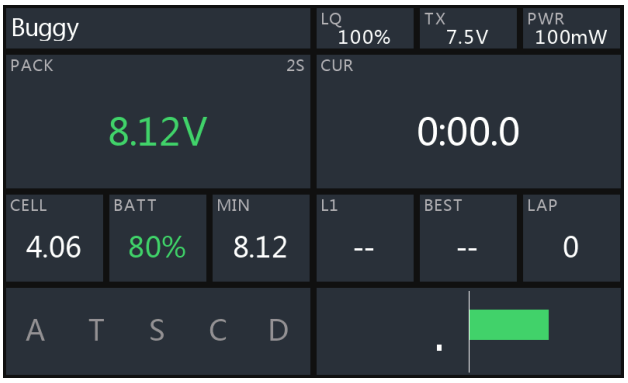

*Drive aids: Aids Switch at the top (aids OFF by default)*




*Battery: alarm, Pack Capacity, Fuel Callouts*

The Brake / ABS and Drive aids settings are all OFF or gentle by default. Leave them there for now: see *About this version* on page 2.

# 8. Reading the dashboard




*On throttle: the bar fills green to the right*




*On the brake: the bar fills red*

| Area | Shows |
|---|---|
| Header (full screen) | Model name, link quality (LQ), radio battery (TX) and transmit power (PWR). |
| PACK | Car pack voltage, with the detected cell count (for example 2S) in the corner. Green above 50%, amber below 50%, red below 20%. |
| CELL / BATT / MIN | Voltage per cell, fuel left (see below), and the lowest pack voltage this session. |
| CUR / L1 / BEST / LAP | Lap timing: current lap, last lap, best lap and lap count. The lap timer arrives in a later version. |
| A T S C D | Drive aid letters: in the aid's own colour while it's working, white when it's armed, grey when it's OFF. Struck through when the Aids Switch has turned the aids off. |
| Live bar | Throttle (green) or brake (red) from the centre line. The small mark below shows the wheel position. |

Telemetry values show **--** until the receiver sends them.

## Fuel (BATT)

| Source | Used when | How it works |
|---|---|---|
| Fuel sensor | The car sends a fuel sensor (for example **Fuel** from the ESC) | Shown as sent. Trackside always prefers it. |
| mAh used | Pack Capacity is set and the car sends mAh used | Starts from the resting voltage, then counts down by mAh used. |
| Voltage | Otherwise | Read from the resting voltage whenever you've been off the throttle for a second. On throttle it holds, and once you've driven it never goes back up. |

For the mAh and voltage sources Trackside creates a telemetry sensor called **Pack Fuel**, so you can use it in logic switches, alerts or other widgets. Check setup shows which source is in use.

## Spoken alerts

| You hear | When |
|---|---|
| Fuel 50 percent (30, 20, 10) | Fuel passes each level, once per pack. Fuel Callouts on the Battery page. |
| Battery low, 3.4 volts + vibration | Voltage Alarm is ON and the resting cell voltage stays under Alarm Cell Voltage for 5 seconds (readings on throttle don't count). Repeats every 30 seconds. |
| Aid test on / Aid test off | The Aid Test on Check setup starts or ends. |
| Drive aids on / Drive aids off + short buzz | You flip the Aids Switch. |
| Drive aids off. Trackside error. + vibration | A Trackside error released the aids: the car drives on the plain sticks. |

Alerts use the radio's voice setting (English: default, US or UK). To use your own recording, put a WAV with the same name in **SCRIPTS/trackside.user/audio**.

# 9. Troubleshooting

| Problem | Check |
|---|---|
| No steering or throttle at all | Trackside isn't loaded and the aid lines have no safety switch. Set both aid lines' Active condition to **LS1** (section 4c), then install Trackside. Also check the Steering and Throttle mixes still use Analogs > Steering and Analogs > Throttle as their source. |
| Throttle or steering moves only a little | Update Trackside to the latest version. Early test builds scaled the aid output wrongly. |
| The aid mix has no effect | Run **Check setup** with the Aid Test (section 5): it says what is wrong. Set the Free mix's **Output** to the channel (CH2 or CH1), check that Operation is **Multiply**, and that the aid line sits **below** its stick mix. LS1 must be green (Model > Logic switches). |
| The aids stopped working after reinstalling Trackside | If the model was changed while Trackside was missing (even a trim click), Ethos clears its Lua links. The car still drives normally. Open each aid line and pick its Lua source again. LS1 and the Trackside sensor are kept. |
| No Lua category when picking a source | Turn on Throttle Aid and Steer Aid in **Model > Lua** (section 3). This is needed once per model. |
| Trackside isn't in the widget list | Check the folder is **SCRIPTS/trackside** with **main.lua** inside, then restart the radio. |
| The dashboard shows -- everywhere | No telemetry yet. Values appear once the receiver is bound and sending telemetry. |
| The bar turns red when I accelerate | Change **Brake Direction** on the tool's Inputs page. |